

Companion LXXI: Hilbert-Space Embeddings of Persistence Diagrams in the Relaxation-Driven Cyclic Cosmology

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April 2026

Abstract

Hilbert-space embeddings of persistence diagrams provide a linearized representation of topological structure, enabling inner-product computations, kernel methods, and spectral analyses. In weak-lensing analyses, such embeddings capture how connected components, loops, and void-like structures contribute to the geometry of topological feature space. In the standard cosmological model, these embeddings reflect the nonlinear matter distribution and the projection of large-scale structure. In the Relaxation-Driven Cyclic Cosmology (RDCC), the scale-dependent evolution of the gravitational potential modifies the Hilbert-space geometry of persistence diagrams in a characteristic way. This Companion develops a continuous description of Hilbert-space embeddings in RDCC, deriving how the relaxation scale influences linearized topology, kernel structure, and the observational signatures in weak-lensing surveys. The resulting framework provides a distinctive probe of the RDCC gravitational field through linearized topological geometry.

1 Introduction

Persistence diagrams encode the birth and death of topological features in weak-lensing convergence fields. While wavelet transforms provide a multi-scale representation of topological fluctuations, Hilbert-space embeddings offer a linearized representation that enables inner-product computations, kernel methods, and spectral analyses.

In the standard cosmological model, the Hilbert-space geometry reflects the nonlinear matter distribution and the projection of large-scale structure. In RDCC, the gravitational potential evolves differently. The relaxation mechanism suppresses the decay of small-scale modes and enhances the coherence of large-scale modes. This modifies the Hilbert-space geometry of persistence diagrams in a scale-dependent manner, producing a distinctive signature in embedding analyses.

2 Hilbert-Space Embeddings of Persistence Diagrams

A persistence diagram D can be embedded into a reproducing kernel Hilbert space (RKHS) $\mathcal{H}$ via

$$\Phi(D) = \sum_{(b,d) \in D} w(b,d) k(\cdot, (b,d)), \quad (1)$$

where k is a positive-definite kernel and $w(b,d)$ is a weight function. The inner product becomes

$$\langle \Phi(D_1), \Phi(D_2) \rangle_{\mathcal{H}} = \sum_{x \in D_1} \sum_{y \in D_2} w(x)w(y)k(x, y). \quad (2)$$

3 RDCC Modification of Persistence Diagrams

In RDCC, the convergence evolves as

$$\kappa_{\text{RDCC}}(\ell) = \kappa(\ell) \left[1 - \chi_{\kappa} \left(\frac{\ell}{\ell_{\text{rel}}} \right)^{\alpha} \right]. \quad (3)$$

This modifies the birth and death thresholds of topological features:

$$b_{\text{RDCC}} = b [1 - \chi_b(\ell/\ell_{\text{rel}})^{\alpha}], \quad (4)$$

$$d_{\text{RDCC}} = d [1 - \chi_d(\ell/\ell_{\text{rel}})^{\alpha}]. \quad (5)$$

Thus the RDCC embedding becomes

$$\Phi_{\text{RDCC}}(D) = \Phi(D) \left[1 - \chi_{\Phi} \left(\frac{\ell}{\ell_{\text{rel}}} \right)^{\alpha} \right]. \quad (6)$$

4 Kernel Structure in RDCC

The RDCC-modified kernel is

$$k_{\text{RDCC}}(x, y) = k(x, y) \left[1 - \chi_k \left(\frac{\ell}{\ell_{\text{rel}}} \right)^{\alpha} \right]. \quad (7)$$

This modification reflects the relaxation-driven change in linearized topology.

5 Linearized Topological Signatures of RDCC

The relaxation mechanism enhances the coherence of large-scale modes. Hilbert-space embeddings therefore exhibit:

- enhanced inner products for large-scale topological features,
- suppressed inner products for small-scale features,
- a characteristic turnover near ℓ_{rel} ,
- modified kernel eigenvalue spectra,
- altered alignment of embedded persistence lifetimes.

These signatures provide a direct probe of the RDCC gravitational field through linearized topological geometry.

6 Conclusion

Hilbert-space embeddings of persistence diagrams in the Relaxation-Driven Cyclic Cosmology reflect the scale-dependent evolution of the gravitational potential and the nonlinear matter distribution. The relaxation mechanism modifies the linearized topology in a scale-dependent manner, producing distinctive signatures in weak-lensing surveys. These effects provide a direct observational window into the relaxation scale and the temporal structure of the RDCC gravitational field. The present Companion establishes the theoretical foundation for future studies of kernel topology, nonlinear structure, and the interplay between light and gravity in RDCC.

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